



# SAFETY DATA SHEET

According to WHS Regulations

Revision date 18-Apr-2025

Revision Number 4.1

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### Product identifier

Product Name Quick Start Bradford Reagent, 1X

Catalogue Number(s) 5000205, 5000205EDU

### Other means of identification

Proper shipping name PHOSPHORIC ACID, SOLUTION

UN number or ID number UN1805

Pure substance/mixture Mixture

### Recommended use of the chemical and restrictions on use

Recommended use Laboratory chemicals

Uses advised against No information available

### Details of manufacturer or importer

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Pty Ltd  
u1A, 62 Ferndell Street,  
South Granville NSW 2142  
Australia

For further information, please contact

Technical Service +61 2 9914 2800 or 1800 224 354  
sales.australia@bio-rad.com

### Emergency telephone number

## SECTION 2: Hazards identification

### GHS Classification

|                                                  |                     |
|--------------------------------------------------|---------------------|
| Corrosive to metals                              | Category 1 - (H290) |
| Flammable liquids                                | Category 4 - (H227) |
| Acute toxicity - Oral                            | Category 4 - (H302) |
| Serious eye damage/eye irritation                | Category 1 - (H318) |
| Specific target organ toxicity — single exposure | Category 1 - (H370) |

### Label elements

Exclamation mark  
Health hazard  
Corrosion

**Signal word**

Danger

**Hazard statements**

H290 - May be corrosive to metals

H227 - Combustible liquid

H302 - Harmful if swallowed

H318 - Causes serious eye damage

H370 - Causes damage to organs

**Precautionary Statements - Prevention**

Wash face, hands and any exposed skin thoroughly after handling

Do not eat, drink or smoke when using this product

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

Wear protective gloves/protective clothing/eye protection/face protection

Keep only in original packaging

**Precautionary Statements - Response**

IF exposed or concerned: Call a POISON CENTER or doctor

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

Immediately call a POISON CENTRE or doctor

IF SWALLOWED: Call a POISON CENTER or doctor if you feel unwell

Rinse mouth

In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

Absorb spillage to prevent material damage

**Precautionary Statements - Storage**

Store in corrosion resistant container with a resistant inner liner

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other hazards which do not result in classification**

No information available.

**SECTION 3: Composition/information on ingredients****Substance**

Not applicable

**Mixture**

| Chemical name             | CAS No.     | Weight-% |
|---------------------------|-------------|----------|
| Phosphoric acid           | 7664-38-2   | 5 - 10   |
| Methanol                  | 67-56-1     | 5 - 10   |
| Non-hazardous ingredients | Proprietary | Balance  |

**SECTION 4: First aid measures****Description of first aid measures**

|                                           |                                                                                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b>                     | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                              |
| <b>Emergency telephone number</b>         | Poisons Information Centre, Australia: 13 11 26<br>Poisons Information Centre, New Zealand: 0800 764 766                                                                                                                                                                       |
| <b>Inhalation</b>                         | Remove to fresh air. Get medical attention immediately if symptoms occur. IF exposed or concerned: Get medical advice/attention.                                                                                                                                               |
| <b>Eye contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Keep eye wide open while rinsing. Do not rub affected area. Get immediate medical advice/attention.               |
| <b>Skin contact</b>                       | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. If symptoms persist, call a doctor.                                                                                                                                      |
| <b>Ingestion</b>                          | Clean mouth with water and drink afterwards plenty of water. Never give anything by mouth to an unconscious person. Do NOT induce vomiting. Call a doctor.                                                                                                                     |
| <b>Self-protection of the first aider</b> | Avoid contact with skin, eyes or clothing. Wear personal protective clothing (see section 8). Remove all sources of ignition. Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

**Most important symptoms and effects, both acute and delayed**

**Symptoms** Burning sensation. Prolonged contact may cause redness and irritation.

**Indication of any immediate medical attention and special treatment needed**

**Note to doctors** Treat symptomatically.

**SECTION 5: Firefighting measures****Suitable Extinguishing Media**

**Suitable Extinguishing Media** Dry chemical. Carbon dioxide (CO<sub>2</sub>). Water spray. Alcohol resistant foam.

**Unsuitable extinguishing media** No information available.

**Specific hazards arising from the chemical**

**Specific hazards arising from the chemical** Keep product and empty container away from heat and sources of ignition. In the event of fire, cool tanks with water spray.

**Special protective actions for fire-fighters**

**Special protective equipment for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

**Hazchem code** 2R

**SECTION 6: Accidental release measures****Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. See section 8 for more information. Take precautionary measures against static discharges. Do not touch or walk

through spilled material.

#### Other information

Refer to protective measures listed in Sections 7 and 8.

#### For emergency responders

Use personal protection recommended in Section 8.

#### Environmental precautions

#### Environmental precautions

Prevent further leakage or spillage if safe to do so. Refer to protective measures listed in Sections 7 and 8.

#### Methods and material for containment and cleaning up

#### Methods for containment

Stop leak if you can do it without risk. Do not touch or walk through spilled material. Dyke far ahead of liquid spill for later disposal.

#### Methods for cleaning up

Take precautionary measures against static discharges. Dam up. Soak up with inert absorbent material. Pick up and transfer to properly labelled containers.

#### Precautions to prevent secondary hazards

#### Prevention of secondary hazards

Clean contaminated objects and areas thoroughly observing environmental regulations.

### SECTION 7: Handling and storage

#### Precautions for safe handling

#### Advice on safe handling

Handle in accordance with good industrial hygiene and safety practice. Do not breathe dust/fume/gas/mist/vapours/spray. Use personal protection equipment. Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. Take precautionary measures against static discharges. Use with local exhaust ventilation. Avoid contact with skin, eyes or clothing. Do not eat, drink or smoke when using this product.

#### General hygiene considerations

Wear suitable gloves and eye/face protection. Do not eat, drink or smoke when using this product. Regular cleaning of equipment, work area and clothing is recommended. Contaminated work clothing should not be allowed out of the workplace. Wash hands before breaks and immediately after handling the product. Avoid contact with skin, eyes or clothing.

#### Conditions for safe storage, including any incompatibilities

#### Storage Conditions

Keep containers tightly closed in a dry, cool and well-ventilated place. Protect from moisture. Store locked up. Keep out of the reach of children. Store away from other materials. Keep away from heat, sparks, flame and other sources of ignition (i.e., pilot lights, electric motors and static electricity). Keep in properly labelled containers. Store in accordance with the particular national regulations. Store in accordance with local regulations.

#### Incompatible materials

Oxidising agent. Strong acids. Strong bases.

### SECTION 8: Exposure controls/personal protection

#### Working area parameters, subject to mandatory control (MAC or TSEL)

#### Exposure Limits

| Chemical name                | Australia                                                 | ACGIH TLV                                             |
|------------------------------|-----------------------------------------------------------|-------------------------------------------------------|
| Phosphoric acid<br>7664-38-2 | TWA: 1 mg/m <sup>3</sup> ;<br>STEL: 3 mg/m <sup>3</sup> ; | TWA: 1 mg/m <sup>3</sup><br>STEL: 3 mg/m <sup>3</sup> |

|                     |                                                                                                  |                                      |
|---------------------|--------------------------------------------------------------------------------------------------|--------------------------------------|
| Methanol<br>67-56-1 | TWA: 200 ppm;<br>TWA: 262 mg/m <sup>3</sup> ;<br>STEL: 250 ppm;<br>STEL: 328 mg/m <sup>3</sup> ; | TWA: 200 ppm<br>STEL: 250 ppm<br>pSk |
|---------------------|--------------------------------------------------------------------------------------------------|--------------------------------------|

**Biological occupational exposure limits**

| Chemical name       | Australia | ACGIH                                     |
|---------------------|-----------|-------------------------------------------|
| Methanol<br>67-56-1 | -         | 15 mg/L - urine (Methanol) - end of shift |

**Appropriate engineering controls**

|                      |                                                     |
|----------------------|-----------------------------------------------------|
| Engineering controls | Showers<br>Eyewash stations<br>Ventilation systems. |
|----------------------|-----------------------------------------------------|

**Individual protection measures, such as personal protective equipment**

|                                 |                                                                                                                                                                          |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Eye/face protection             | Tight sealing safety goggles.                                                                                                                                            |
| Skin and body protection        | Wear suitable protective clothing.                                                                                                                                       |
| Hand protection                 | Wear suitable gloves.                                                                                                                                                    |
| Respiratory protection          | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |
| Environmental exposure controls | No information available.                                                                                                                                                |

**SECTION 9: Physical and chemical properties****Information on basic physical and chemical properties**

|                 |                          |
|-----------------|--------------------------|
| Physical state  | Liquid                   |
| Appearance      | aqueous solution         |
| Colour          | light blue               |
| Odour           | Alcohol.                 |
| Odour threshold | No information available |

| Property                                | Values            | Remarks • Method |
|-----------------------------------------|-------------------|------------------|
| pH                                      |                   | None known       |
| Melting point / freezing point          | No data available | None known       |
| Initial boiling point and boiling range | 64.7 °C           |                  |
| Flash point                             | 65 °C             |                  |
| Evaporation rate                        | No data available | None known       |
| Flammability                            | No data available | None known       |
| Flammability Limit in Air               |                   | None known       |
| Upper flammability or explosive limits  | No data available |                  |
| Lower flammability or explosive limits  | No data available |                  |
| Vapour pressure                         | No data available | None known       |
| Relative vapour density                 | No data available | None known       |
| Relative density                        | No data available | None known       |
| Water solubility                        | Miscible in water |                  |
| Solubility(ies)                         | No data available | None known       |
| Partition coefficient                   | No data available | None known       |

|                           |                   |            |
|---------------------------|-------------------|------------|
| Autoignition temperature  | No data available | None known |
| Decomposition temperature |                   | None known |
| Kinematic viscosity       | No data available | None known |
| Dynamic viscosity         | No data available | None known |
| Explosive properties      | Not applicable    |            |
| Oxidising properties      | Not applicable    |            |
| <b>Other information</b>  |                   |            |
| Molecular weight          | Not applicable    |            |
| VOC content               | Not applicable    |            |

## SECTION 10: Stability and reactivity

### Reactivity

**Reactivity** No information available.

### Chemical stability

**Stability** Stable under normal conditions.

### Explosion data

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** Yes.

### Possibility of hazardous reactions

**Possibility of hazardous reactions** None under normal processing.

### Conditions to avoid

**Conditions to avoid** Exposure to air or moisture over prolonged periods. Heat, flames and sparks.

### Incompatible materials

**Incompatible materials** Oxidising agent. Strong acids. Strong bases.

### Hazardous decomposition products

**Hazardous decomposition products** None known based on information supplied.

## SECTION 11: Toxicological information

### Acute toxicity

### Information on likely routes of exposure

#### Product Information

|                     |                                                                                                                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>   | Specific test data for the substance or mixture is not available.                                                                                                                 |
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available. Causes serious eye damage. May cause irreversible damage to eyes.                                               |
| <b>Skin contact</b> | Specific test data for the substance or mixture is not available. Causes mild skin irritation.                                                                                    |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhoea. Harmful if swallowed (based on |

components)

**Symptoms**

Redness. Burning. May cause blindness. Prolonged contact may cause redness and irritation.

**Numerical measures of toxicity - Product Information**

No information available

The following values are calculated based on chapter 3.1 of the GHS document

|                               |                |
|-------------------------------|----------------|
| ATEmix (oral)                 | 1,800.00 mg/kg |
| ATEmix (dermal)               | 5,058.50 mg/kg |
| ATEmix (inhalation-vapour)    | 834.00 mg/l    |
| ATEmix (inhalation-dust/mist) | 10.00 mg/l     |

**Component Information**

| Chemical name   | Oral LD50            | Dermal LD50              | Inhalation LC50                      |
|-----------------|----------------------|--------------------------|--------------------------------------|
| Phosphoric acid | = 1530 mg/kg ( Rat ) | = 2740 mg/kg ( Rabbit )  | = 3846 mg/m <sup>3</sup> ( Rat ) 1 h |
| Methanol        | = 6200 mg/kg ( Rat ) | = 15840 mg/kg ( Rabbit ) | = 22500 ppm ( Rat ) 8 h              |

See section 16 for terms and abbreviations

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                          |                                                                                    |
|------------------------------------------|------------------------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>         | May cause skin irritation. Classification based on data available for ingredients. |
| <b>Serious eye damage/eye irritation</b> | Classification based on data available for ingredients. Causes burns.              |
| <b>Respiratory or skin sensitisation</b> | Based on available data, the classification criteria are not met.                  |
| <b>Germ cell mutagenicity</b>            | Based on available data, the classification criteria are not met.                  |
| <b>Carcinogenicity</b>                   | Based on available data, the classification criteria are not met.                  |
| <b>Reproductive toxicity</b>             | Based on available data, the classification criteria are not met.                  |
| <b>STOT - single exposure</b>            | Causes damage to organs if swallowed.                                              |
| <b>STOT - repeated exposure</b>          | Based on available data, the classification criteria are not met.                  |
| <b>Aspiration hazard</b>                 | Based on available data, the classification criteria are not met.                  |

**SECTION 12: Ecological information****Ecotoxicity**

**Ecotoxicity** Harmful to aquatic life.

| Chemical name | Algae/aquatic plants | Fish                                                                                                                                    | Toxicity to microorganisms | Crustacea |
|---------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------|
| Methanol      | -                    | LC50: =28200mg/L (96h, Pimephales promelas)<br>LC50: >100mg/L (96h, Pimephales promelas)<br>LC50: 19500 - 20700mg/L (96h, Oncorhynchus) | -                          | -         |

|  |  |                                                                                                                        |  |  |
|--|--|------------------------------------------------------------------------------------------------------------------------|--|--|
|  |  | mykiss)<br>LC50: 18 - 20mL/L (96h,<br>Oncorhynchus mykiss)<br>LC50: 13500 - 17600mg/L<br>(96h, Lepomis<br>macrochirus) |  |  |
|--|--|------------------------------------------------------------------------------------------------------------------------|--|--|

**Persistence and degradability**

**Persistence and degradability** No information available.

**Bioaccumulative potential**

**Bioaccumulation** There is no data for this product.

**Component Information**

| Chemical name   | Partition coefficient |
|-----------------|-----------------------|
| Phosphoric acid | -0.9                  |
| Methanol        | -0.77                 |

**Mobility**

**Mobility in soil** No information available.

**Mobility** No information available.

**Other adverse effects**

**Other adverse effects** No information available.

**SECTION 13: Disposal considerations****Disposal methods**

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

**SECTION 14: Transport information****ADG**

**UN number or ID number** UN1805  
**Proper shipping name** PHOSPHORIC ACID, SOLUTION  
**Transport hazard class(es)** 8  
**Packing group** III  
**Special Provisions** 223  
**Description** UN1805, PHOSPHORIC ACID, SOLUTION, 8, III  
**Hazchem code** 2R

**IATA**

**UN number or ID number** UN1805  
**UN proper shipping name** Phosphoric acid, solution  
**Transport hazard class(es)** 8  
**Packing group** III  
**ERG Code** 8L



|                                   |                                           |
|-----------------------------------|-------------------------------------------|
| <b>Special Provisions</b>         | A3, A803                                  |
| <b>Description</b>                | UN1805, Phosphoric acid, solution, 8, III |
| <b>IMDG</b>                       | Not regulated                             |
| <b>UN number or ID number</b>     | UN1805                                    |
| <b>UN proper shipping name</b>    | PHOSPHORIC ACID SOLUTION                  |
| <b>Transport hazard class(es)</b> | 8                                         |
| <b>Packing group</b>              | III                                       |
| <b>EmS-No.</b>                    | F-A, S-B                                  |
| <b>Special Provisions</b>         | 223                                       |
| <b>Marine pollutant</b>           | NP                                        |
| <b>Description</b>                | UN1805, PHOSPHORIC ACID SOLUTION, 8, III  |

**Transport in bulk according to Annex II of MARPOL and the IBC Code**

No information available

**SECTION 15: Regulatory information****Safety, health and environmental regulations/legislation specific for the substance or mixture****National regulations****Australia**

See section 8 for national exposure control parameters

**Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP)**

Classified as a scheduled poison according to the Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP)

**Poison Schedule Number** 6**National pollutant inventory**

Subject to reporting requirement

| Chemical name               | National pollutant inventory         |
|-----------------------------|--------------------------------------|
| Phosphoric acid - 7664-38-2 | 10 tonne/yr Threshold category 1     |
| Methanol - 67-56-1          | 10 tonne/yr Threshold category 1 VOC |

**Banned and/or restricted**

This product contains one or more substance(s) subject to prohibition, authorisation or restriction. Verify that requirements related to using, handling, and storing substances subject to prohibition, authorisation or restriction are met.

| Chemical name      | Carcinogen | Restricted substance                                   |
|--------------------|------------|--------------------------------------------------------|
| Methanol - 67-56-1 |            | For spray painting at a concentration of >1% by volume |

**International Inventories**

Contact supplier for inventory compliance status

**International Regulations****The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable**The Stockholm Convention on Persistent Organic Pollutants** Not applicable**The Rotterdam Convention** Not applicable**SECTION 16: Other information**

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety

**Revision date** 18-Apr-2025

**Revision Note** SDS sections updated. 14.

**Key or legend to abbreviations and acronyms used in the safety data sheet**

Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
Environmental Protection Agency  
Acute Exposure Guideline Level(s) (AEGl(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
U.S. National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**